

Digital Forensic Court-Ready Summary

[REDACTED]

[REDACTED]

Somsouk Thepphavong [REDACTED]

[REDACTED]



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Executive Summary

This report presents a court-ready forensic evidence summary regarding the suspected unauthorised disclosure of confidential data ("Seed Files") by Mr. Iaman Informant. The forensic investigation was conducted between the timeframe of 23rd March 2015 and 25th March 2015.

Forensic investigation revealed that Mr. Iaman Informant communicated via email with an external individual, discussing the exchange of confidential information in return for payment. Evidence recovered from the Windows registry hive, Master File Table (\$MFT), removable media, browser history and file-system artifacts confirmed that 17 Seeds Files were extracted from the company's secure network drive and copied onto an unauthorised USB device (RM#2). Consequently, the suspect executed two anti-forensic applications, CCleaner and Eraser, to delete and obscure system activity, responsible for the deletion of more than 800 files from the machine.

Hash verification, chain of custody documentation, and adherence to accepted digital forensic standards confirm the reliability and integrity of the evidence. In accordance with Rule 402, the evidence presented is relevant and directly supports the allegation of corporate espionage. Applying Rule 901, the digital artifacts have been properly authenticated. Under Rule 502, privileged communications were not included.

Based on the evidence gathered, the probability that these findings accurately attribute the data-leakage actions to Mr Informant exceeds 95%. Forensic evidence supports the conclusion that intentional, unauthorised data disclosure occurred to an unauthorised third party.

1. Introduction

This report presents a forensic evidence summary on the investigation into a suspected corporate data leakage incident that occurred between March 23, 2015, and March 25, 2015. The investigation was initiated following the prior identification of suspicious removable media and the suspected unauthorised disclosure of confidential company information, specifically proprietary "Seed Files".

The investigation focused on digital data and artifacts with Mr. Iaman Informant, the registered owner of the office computer under examination.

The purpose of this document is to verify the findings of the forensic investigation and summarise the evidence for presentation in the Australian Court of Law. The report objectives are

- (a) Outline the methodology used to identify, collect and analyze relevant digital evidence
- (b) Present the facts related to exfiltration and cover-up

This report is presented to the Court to aid the trier of fact in establishing the root cause and the culpability of the accused.

1.1. Statement of Compliance

It is understood that the duty of the forensic investigator and expert witness is to provide evidence and analysis that is objective, unbiased and within the scope of the examiner's knowledge and expertise. The examiner will inform the Court and all parties immediately if any factors arise that could compromise the objectivity or independence of the findings presented in this report.

2. Title and Scope of Work

2.1. Engagement And Authority

The examination detailed in this report was requested by the Australian Court of Law. Following the referral of seized digital evidence for forensic investigation.

Examiner:

[REDACTED]

[REDACTED]

Somsouk Thepphavong – Lead Investigator

2.2. Purpose of Scope

This report presents the findings of a digital forensic examination concerning the suspected unauthorised transfer of confidential data by Mr. Iaman Informant. The examination was conducted for academic purposes and focused on the analysis of the suspect's workstation image and removable media devices (RM#1, RM#2, RM#3).

The scope of work included acquisition, verification and analysis of non-volatile artefacts; validation of relevant timestamps; and correlation of user activity to identify potential data-leakage events. No live memory or network capture was obtained.

In producing this court-ready report, the forensic team followed the rules of evidence 402 and 502 set by US law. As Rule 402 of the Federal Rules of Evidence states, only relevant evidence is generally admissible in court (Federal Rules of Evidence, 2024, p. 4). Furthermore, the forensic team adhere to Rule 502 of the Federal Rules of Evidence, which states that communication between an attorney and a client is legally considered a secret. The methodology included reasonable steps (such as keyword filtering) to segregate potentially privileged communications identified during the analysis (Federal Rules of Evidence, 2024, p. 10).

3. Chain of Custody

Date/Time	Released By	Signature	Received By	Signature	Purpose	Location

4. Methodology

4.1. Item Submitted for Investigation

Personal Computer

- Windows 7 Ultimate (SP1) Registered Owner – Iaman Informant
- PC Name – Informant-PC Processor Architecture AMD64
- Install Date – 1427034866 in Unix epoch
 - Convert to UTC – 2015-03-22 07:14:26.
- The time zone of the computer – Eastern Standard Time (UTC -5)

Removable Media #1

- This is the company's authorised external storage device, permitted to store the company's highly classified internal information.
- Anything considered to be classified information will be referred to as "Seed File".
- Model – SanDisk Corp Cruzer Fit
- Device ID – 4C530012450531101593

Removable Media #2

- This is the submitted item confiscated by the security guard from Mr Iaman Informant.
- Model – SanDisk Corp Cruzer Fit
- Device ID – 4C530012550531106501

Removable Media #3

- An external CD-ROM was confiscated at the security checkpoint.

The evidence collected consisted of file images. The integrity of these images was confirmed by calculating and comparing SHA1 hash values against official control values, thereby ensuring that the content was not altered or tampered with.

4.2. Digital Forensic Workflow

- **1. Evidence Acquisition:**
 - Bit-by-bit imaging using FTK Imager
- **2. Integrity Verification:**
 - SHA-1 hashes of images were matched against the original hash value
- **3. Artifact Extraction:**
 - RegistryExplorer (registry hives)
 - MFTECmd (MFT timestamps)
 - LECmd (LNK files)
 - EvtxECmd (Event Logs)
 - PECmd (Prefetch)
- **4. Timeline Correlation:**
 - Forensic timestamps were converted to UTC
- **5. Privileged Data Exclusion:**
 - In compliance with Rule 502, privilege-related content was not included in.
- **6. Relevance Assessment:**
 - Each artifact was evaluated under Rule 402 to ensure probative value.

5. Chronological Order

On the 23rd of March 2015, at 17:30 UTC, the first email was sent to Mr. Iaman (iaman-informat@nist.gov.ost) from Mr. Conspirator (spy.conspirator@nist.gov). The email conversation was conducted using Outlook. The context of an email indicates an offer from Mr. Conspirator of high payment in exchange for confidential information.

On the 23rd of March 2015, at 20:25 UTC, a network artifact was created when an individual accessed the company's Secured Network Drive. Stored in the IP address of 10.11.11.128, where "Seed Files" were stored.

On the 24th of March 2015, at 19:38 UTC, the USB drive RM#2 was inserted to extract "Seed Files" to its physical media. 17 out of 22 seed files were retrieved, verified using their hash values, and confirmed to be identical to the confidential information via running the hash command from PowerShell.

On the 24th of March, at 21:05 UTC, both parties agreed to meet the day after for the exchange of physical storage media devices via email.

On the 25th of March, at 11:08 UTC, both CCleaner and the Eraser executable program were installed on the personal computers.

Programs	Description
Ccleaner	File-cleaning tools for performance with complete file deletion features.
Eraser	File-cleaning tools capable of overwriting the disk multiple times to ensure the complete removal of sensitive data from the hard drive.

Eric Zimmerman's tool, PECmd, can parse the Prefetch file and extract information about any running process. It was confirmed that both programs were installed.

- Eraser.EXE was executed on 25th of March 2015, at 15:13:30 UTC.
- CCLEANER.EXE was executed on 25th of March 2015, at 15:15:50 UTC.

See Appendix A for a visualized timeline diagram for better understanding.

6. Results and Root Cause

The evidence demonstrates that Mr. Iaman Informant:

1. Communicated with an external actor (Mr. Spy Conspirator), intending to leak Seed Files
2. Accessed the company's secure data repository
3. Inserted an unauthorized USB (RM#2)
4. Extracted 17 Seed Files to an unauthorized USB (RM#2)
5. Installed and executed anti-forensic tools to conceal evidence
6. Employed timestomping anti-forensic technique to manipulate file timestamps.

This sequence indicates a clear intent, action and attempted concealment.

7. Conclusion

This forensic investigation successfully identified the root cause of the data leakage incident. The compilation of gathered evidence established a clear timeline demonstrating the culpability of Mr. Iaman Informant.

Mr. Iaman's action was considered as an act of corporate espionage. Combined with email artefacts, followed by the physical extraction of "Seed Files" onto the unauthorized RM#2. Additionally, the execution of CCleaner and Eraser, together with Timestomping, is an anti-forensic technique. This sequence of events indicates direct evidence of intention to conceal digital crime.

Given the multiple, the probability that these findings are accurate and linked to Mr. Iaman Informant is estimated at 95% or higher. The findings would be presented in court using a diagram from **Appendix A**, allowing the Court to clearly understand how the technical evidence proves both the act and the intent of the accused.

References

U.S. House of Representatives, Committee on the Judiciary. (2024). *Federal Rules of Evidence* (December 1, 2024 ed). U.S. Government Publishing Office.

Appendix A

Chronological Order Diagram

